Pandelis Zembashis

CTO | ML Platform Engineering | Inference Infrastructure & Reliability

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EXECUTIVE SUMMARY

CTO and hands-on platform engineer building production AI systems where model capability depends on reliable infrastructure. At Cosine, built and led the runtime, secure execution, orchestration, observability, and deployment foundations behind an AI coding platform spanning local clients, remote Kubernetes workers, enterprise VPCs, on-prem, and air-gapped environments. Strong fit for Iso's ML Ops role through production ML workload operation, inference-adjacent service design, reliability-first engineering, GPU/HPC-aware deployment, and the ability to work between research, product, and operations teams.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – May 2026

Applied AI lab building coding-agent models, runtimes, sandboxed compute, and enterprise deployment infrastructure.

- Led product and engineering for Cosine's AI coding platform, setting direction across runtime architecture, cloud agents, secure compute, developer experience, reliability, observability, and enterprise deployment.
- Built the remote execution substrate for long-running AI workloads: Kubernetes-scheduled workers, gVisor/runsc isolation, ephemeral sessions, WebSocket control planes, secure shell/tool execution, and per-tenant execution boundaries.
- Designed reliability patterns for model-driven workloads, including checkpointing, deterministic replay, event logs, state propagation, cancellation, recovery, and debugging hooks for production failure analysis.
- Shipped one Go runtime across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, keeping execution semantics stable across local, embedded, and cloud-backed environments.
- Built deployment paths for managed cloud, private VPC, on-prem, and air-gapped customers where customer code, execution logs, model artifacts, and tenant data need to remain isolated, observable, and auditable.
- Built workload evidence around the runtime: replayable traces, tests, LSP diagnostics, checkpoints, reviews, parallel sandboxed runs, and feedback loops for benchmarking model/tool behavior under real developer workloads.
- Helped position Cosine as an AI lab and sovereign coding platform, including public coverage of benchmark-leading models, 38+ language support, air-gapped deployment, and 500,000 AIRR GPU hours on Isambard-AI.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound automation systems that processed millions of email chains and billions of tokens, with event-driven architecture, data pipelines, worker orchestration, and operational monitoring.
- Established engineering cadence, incident response habits, onboarding, and operating rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving iteration across web, operations, and marketplace workflows.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

SELECTED PUBLIC SIGNALS

Tech.eu on Cosine sovereign AI: Coverage of Cosine as an AI lab and product company building models, agents, infrastructure, and sovereign deployment paths.
Why Cosine built its own harness: Public writing on building internal runtime, orchestration, tool-use, and reliability infrastructure for AI software engineering systems.
Cosine Swarm: Public writing on long-horizon agents, parallel execution, orchestration, and the infrastructure needed to run autonomous workloads.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

ML Platform & Runtime Infrastructure: Production AI workloads, inference-adjacent services, agent runtimes, tool execution, replay, evaluation signals
Kubernetes & Secure Compute: Kubernetes workers, gVisor/runsc, tenant isolation, VPC/on-prem/air-gapped deployment, CI/CD, release hardening
Reliability & Observability: Tracing, event logs, checkpoints, recovery, incident response, benchmarking, workload evidence, operational debugging
Cloud & Systems Engineering: Go, TypeScript, Node.js, Kubernetes/GKE workload patterns, AWS, WebSockets, distributed systems, GraphQL, React
Cross-Functional Leadership: Technical roadmaps, research-product loops, stakeholder alignment, fast-growth engineering teams, platform strategy